

HAMMAD UL EAMAN MIR

+91-7006988121 | erhammadmir@gmail.com | github.com/Hammad-Mir | linkedin.com/in/hammad-mir |
hammad-mir.github.io/portfolio

Machine Learning Researcher | 3+ Years Research Experience | Biomedical Imaging & Applied AI

PROFESSIONAL SUMMARY

Machine learning researcher and engineer spanning biomedical imaging and applied AI: research in multi-modal image registration and weakly supervised histopathology segmentation, followed by production experience building end-to-end computer vision pipelines — from dataset annotation and model development to validation and deployment. Comfortable combining deep learning, representation learning, and classical statistical methods to solve real, data-scarce problems.

EDUCATION

MSc, Data Science (Specialisation in AI)

Newcastle University

Grade: Distinction
Newcastle Upon Tyne, UK

B.Tech, Computer Science and Technology

Islamic University of Science and Technology

CGPA: 8.46/10
Awantipora, India

EXPERIENCE

AI/ML Engineer (Applied Research)

GraphicWeave

Jun 2025 - Present
Srinagar, J&K, India

- Lead the design and development of a computer vision pipeline for structured information extraction from engineering schematics (P&IDs), directing a small team of interns and junior engineers across annotation, model development, and evaluation.
- Re-annotated the full training dataset from scratch (roughly three months) and consolidated 1,100+ inconsistent per-vendor legend entries (20,533 symbols across 4 operator datasets) into 12 standardised equipment categories — unblocking automation that had previously stalled on cross-vendor legend inconsistency.
- Developed the object detection stage using YOLOv11 over a long-tailed symbol vocabulary (250 classes, scaling to 800+), addressing class imbalance through class-aware sampling, copy-paste augmentation, and focal loss; achieved 76% mAP@50–95 (93% mAP@50, 96% precision, 90% recall) on the detection benchmark.
- Trained a consolidated symbol classifier (ConvNeXt-Tiny) reaching 99.5% accuracy on known document styles and ~98% accuracy on a brand-new client's previously unseen drawings with no retraining; extended it with a Siamese/metric-learning network and foundation vision models (DINOv3 ViT-B/16 with LoRA/PEFT) for exact per-document symbol matching.
- Built the downstream stage converting detected symbols and line connections into a Neo4j graph representation of equipment relationships, supporting automated process-safety analysis (AI-assisted HAZOP and LOPA).
- Developed an agentic multimodal retrieval-augmented generation (RAG) system for technical troubleshooting, enabling context-aware querying over manuals and schematics.

Computational Researcher

Peratech Holdco Ltd

Mar 2024 – Jan 2025
Durham, UK

- Led R&D on a behavioural biometric authentication system using high-frequency force-sensing keyboard data; achieved 76% accuracy on the operationally realistic one-class task (genuine user vs unseen impostors), with the closed-set binary task fully separable.
- Developed a dual-pipeline architecture: a C# interface for real-time acquisition (100Hz polling, synchronised logging of keystroke events and 144-channel pressure signals) and a Python ML pipeline for preprocessing, feature engineering, and representation learning.
- Designed and implemented end-to-end data pipelines for high-dimensional time-series data, reducing storage footprint by 100× and improving processing efficiency by 40%.
- Designed controlled user studies capturing over 2,500 samples, incorporating behavioural variability and fatigue-related factors to test robustness of the learned representations.

Research Assistant – Digital Pathology

Newcastle University

Sep 2022 - Dec 2022
Newcastle Upon Tyne, UK

- Conducted a structured survey of multi-modal biomedical image registration methods for breast cancer imaging, comparing transformer-based and CNN-based approaches across H&E, IMC, and LA-ICP-MS modalities; produced 20+ detailed methodological reviews.

- Developed data preprocessing and quality-control workflows for the registration pipeline, and analysed the literature on cross-modal alignment, characterising the robustness limitations and applicability of state-of-the-art techniques.

Research Project – Digital Pathology

May 2022 – Aug 2022

Newcastle University

Newcastle upon Tyne, UK

- Designed and implemented the first transformer-based weakly supervised semantic segmentation framework for histopathology, using DeiT/Swin classifiers with GradCAM and Pixel-Adaptive Refinement to generate pseudo-labels for a DeepLabV3+ segmentation model, trained from image-level labels only.
- Achieved a test mIoU of 0.90 on the WSSS4LUAD benchmark (a top-5 result), with 0.94 IoU on normal tissue — exceeding the previous best reported score of 0.89 — and 0.86 IoU on merged tumour–stroma regions.
- Addressed limited-annotation and gigapixel-scale challenges through patch-based processing of whole-slide images, targeted augmentation, and ablation studies in Python and PyTorch, ensuring reproducibility.

Research Internship – Medical Image Analysis

Jan 2019 – Feb 2019, Oct 2019 – Nov 2019

Indian Institute of Technology (IIT) Mandi

Mandi, Himachal Pradesh, India

- Evaluated deep learning architectures (ResNet, DenseNet) for acute lymphoblastic leukaemia detection, fine-tuning pre-trained models with patch-wise classification to achieve 85% accuracy on the C_NMC 2019 dataset.
- Explored autoencoder-based dimensionality reduction and class-imbalance strategies to improve generalisation on medical imaging datasets.

Undergraduate Final-Year Project – Medical Imaging

Sept 2018 – Aug 2019

Islamic University of Science and Technology (IUST)

Awantipora, J&K, India

- Developed a custom CNN for acute lymphoblastic leukaemia detection, achieving 93% classification accuracy through systematic dataset analysis and benchmarking against baseline methods.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, SQL, PyTorch, TensorFlow, Numpy, Pandas, scikit-learn, C#
- **Machine Learning & Computer Vision:** Supervised & Weakly Supervised Learning, Image Segmentation & Registration, CNNs, Vision Transformers, Foundation Models (DINOv3, LoRA/PEFT), Representation Learning, Model Evaluation, Ablation Studies
- **Biomedical Imaging:** Digital Pathology & Histopathology, Multi-modal Imaging (H&E, IMC, LA-ICP-MS), Patch-wise Learning, Large Image Processing, Learning under Limited Annotations
- **Data & Experimental Pipelines:** Data Preprocessing & QA, Experimental Design & Reproducibility, ETL Pipelines, GPU-based Training Environments
- **Software & Research Tools:** Git, Docker
- **Research & Professional:** Independent project ownership, interdisciplinary collaboration, technical writing and dissemination, cross-functional team leadership